

ECON 4050 - Final Project: Economic Freedom & Economic Growth

Trevor Blum

2026-04-20

Introduction

Economic freedom is not just a concept. It decides whether someone can start a business without having to deal with government red tape, if the money they save will be worth anything in ten years, and if the goods and services they need can move freely across borders. The difference between the most and least economically free countries reflects the difference between markets that work and ones that don't, and between institutions that reward hard work and ones that punish it. Determining what really gives a country economic freedom is more than just an academic exercise. It has real effects on real people.

This intuition brings about the central question of this study. Which elements of economic freedom are most strongly related to economic growth, and which variables contribute most to overall economic freedom? The answer turns out to be more complicated than the conventional wisdom suggests.

This paper looks at how economic freedom relates to per capita GDP across 164 countries between 2000 and 2023, using the Fraser Institute's Economic Freedom of the World index as the main data source. The hypothesis going in was that higher scores across all five component areas (Size of Government, Legal System, Sound Money, Trade Freedom, and Regulation) would go along with higher income levels, since freer markets generally support better incentives for investment and entrepreneurship. Four of the five components bear that out. The fifth one doesn't, and the way it fails raises some uncomfortable questions about what the index is actually measuring.

Literature Review

The Fraser Institute's Economic Freedom of the World index dates back to 1996, when Gwartney, Lawson, and Block first put it together, and it has since become about as close to a standard measurement tool as the field has. It combines 45 indicators into five broad categories: Size of Government, Legal System and Property Rights, Sound Money, Freedom to Trade Internationally, and Regulation. Each are weighted equally in the final score. The research record built on it is substantial, with over 700 peer-reviewed studies published using it, and the general pattern across that literature is that higher-scoring countries tend to be wealthier, grow faster, and have lower poverty rates (Gwartney, Lawson, and Murphy, 2024).

However, not everyone has been satisfied treating the index as a single coherent measure. Heckelman and Stroup (2000) were among the first to break it apart and look at each component on its own, and what they found was uncomfortable: the five dimensions do not all contribute equally to growth, and some, particularly those related to government size and taxation, show weak or even negative associations with economic performance. Their follow-up work (Heckelman and Stroup, 2005) pushed further, showing that the choice of aggregation method can substantially change what the data appears to say, a methodological problem that hasn't been cleanly resolved since.

More recent work keeps turning up the same issue. The Size of Government component in particular shows up repeatedly as negatively and only loosely related to the other four (Heckelman and Stroup, 2005; Kreft and Sobel, 2005), which raises a real question about whether it belongs in the same composite index at all.

When researchers dig into the subcomponents rather than the aggregates, rule of law, monetary stability, and functioning property rights hold up much more reliably as predictors than low government spending or low tax rates do specifically.

This paper works in that tradition. Rather than taking the index at face value, it pulls it apart and asks which institutional dimensions actually drive overall economic freedom scores and which ones are doing something different. The goal is to connect that question directly to economic outcomes, using GDP per capita as an independent check on whatever patterns turn up in the index itself.

Data

Data Sources

The analysis draws on two datasets. The first is the Fraser Institute's EFW dataset, which provides annual country-level scores for overall economic freedom and each of the five component areas, running from 1970 through 2023. The second is the World Bank's World Development Indicators, which provides GDP per capita in constant 2015 US dollars — the main outcome variable for the regression. The two datasets are merged by ISO country code and year to create a single panel dataset covering 164 countries from 2000 to 2023.

Setup

```
library(tidyverse)      # ggplot2, dplyr (select, filter, mutate, etc.)
library(readxl)         # read Excel files
library(broom)          # tidy() and glance() for regression output
library(modelsummary)   # side-by-side regression tables

setwd("C:/Users/apeti/OneDrive/Documents/ECON 4050/c) Final Project")
```

Load Data

```
# Fraser Institute - Economic Freedom of the World Index
fraser_raw <- read_excel(
  "C:/Users/apeti/OneDrive/Documents/ECON 4050/c) Final Project/economic-freedom-of-the-world-2025-master.xlsx",
  sheet = "EFW Panel Dataset"
)

# World Bank - World Development Indicators
wdi_raw <- read_excel(
  "C:/Users/apeti/OneDrive/Documents/ECON 4050/c) Final Project/P_Data_Extract_From_World_Development_Indicators.xlsx",
  sheet = "Data"
)
```

Clean Data

Fraser EFW

The Fraser dataset is renamed and filtered to 2000 onwards, which is where country coverage becomes consistent enough for panel analysis. Each component score is converted to numeric format so R treats them as continuous variables rather than character strings, which is a necessary step to perform prior to the analysis.

```

fraser_clean <- fraser_raw %>%
  rename(
    iso_code = ISO_Code,
    country  = Countries,
    year     = Year,
    efw_total = Summary,
    efw_gov   = `Area 1`,      # Size of Government
    efw_legal = `Area 2`,      # Legal System & Property Rights
    efw_money = `Area 3`,      # Sound Money
    efw_trade = `Area 4`,      # Freedom to Trade Internationally
    efw_reg   = `Area 5`,      # Regulation
    region    = `World Bank Region`,
    income     = `World Bank Current Income Classification, 1990-Present`
  ) %>%
  select(iso_code, country, year, efw_total, efw_gov,
         efw_legal, efw_money, efw_trade, efw_reg, region, income) %>%
  filter(year >= 2000) %>%
  mutate(efw_total = as.numeric(efw_total),
         efw_gov   = as.numeric(efw_gov),
         efw_legal = as.numeric(efw_legal),
         efw_money = as.numeric(efw_money),
         efw_trade = as.numeric(efw_trade),
         efw_reg   = as.numeric(efw_reg))

```

World Bank WDI

The World Bank data needed more work to get into usable shape. It came downloaded in wide format, with each year as its own column, which has two separate data series stacked on top of each other in the same file. This is not how R wants the data to be displayed for any kind of analysis. Getting it into a workable format meant first collapsing all those year columns into a single year variable, then separating the two series out into their own columns so each row represented one country, one year, with one GDP value attached to it.

```

wdi_long <- wdi_raw %>%
  rename(
    country_name = `Country Name`,
    iso_code     = `Country Code`,
    series_name  = `Series Name`,
    series_code  = `Series Code`
  ) %>%
  # pivot_longer() collapses all the year columns (2000, 2001, 2002 etc.)
  # into two columns: one called year_label and one called value.
  # This converts the data from wide format to long format
  # so that each row represents one country-year observation.
  pivot_longer(
    cols      = starts_with("20"),
    names_to  = "year_label",
    values_to = "value"
  ) %>%
  # str_sub() extracts just the first 4 characters from the year label.
  # For example "2000 [YR2000]" becomes just "2000".
  # This gives us a clean numeric year variable we can merge on.
  mutate(year = as.numeric(str_sub(year_label, 1, 4))) %>%
  # as.numeric() converts "." (World Bank missing code) to NA automatically.
  mutate(value = as.numeric(value)) %>%

```

```

select(iso_code, year, series_name, value)

## Warning: There was 1 warning in `mutate()`.
## i In argument: `value = as.numeric(value)`.
## Caused by warning:
## ! NAs introduced by coercion

# pivot_wider() does the opposite of pivot_longer().
# It takes the two series stacked on top of each other
# and spreads them into their own separate columns,
# giving us one row per country-year with gdp_pc and gdp_growth as columns.
# values_fn = mean handles any duplicate observations by averaging them.
wdi_clean <- wdi_long %>%
  pivot_wider(
    names_from = series_name,
    values_from = value,
    values_fn = mean
  ) %>%
  rename(
    gdp_pc = `GDP per capita (constant 2015 US$)`,
    gdp_growth = `GDP per capita growth (annual %)`
  )

```

Merge Datasets

Once both datasets were cleaned, merging them was straightforward. It required matching on ISO country code and year so each row represents one country in one year with both its EFW scores and its GDP figure attached. Any observation missing either GDP per capita or the overall EFW score was subsequently dropped at this stage, since there is nothing useful you can do with an incomplete row in a regression. GDP per capita was then log-transformed before any analysis. The reason for this is practical: GDP per capita ranges from a few hundred dollars in the poorest countries to over seventy thousand in the richest, and that kind of scale difference makes the raw relationship between variables highly non-linear and hard to model cleanly with OLS. Taking the natural log compresses that range and brings the relationship much closer to linear, which is one of the core assumptions the regression depends on. This is standard practice in growth economics for exactly this reason.

```

merged_data <- fraser_clean %>%
  left_join(wdi_clean, by = c("iso_code", "year")) %>%
  filter(!is.na(gdp_pc), !is.na(efw_total)) %>%
  mutate(gdp_pc = as.numeric(gdp_pc),
         log_gdp_pc = log(gdp_pc))

# Quick checks
nrow(merged_data)

## [1] 3859

n_distinct(merged_data$iso_code)

## [1] 164

range(merged_data$year)

## [1] 2000 2023

```

The merged dataset contains 3,859 observations across 164 countries from 2000 to 2023.

Summary Statistics

Before running any formal analysis, it helps to get a sense of what the data actually looks like. The table below shows basic descriptive statistics for the key variables. EFW component scores all run on the same 0 to 10 scale, so they're directly comparable to each other. This is something worth keeping in mind when reading the regression output later. A few things jump out even at this stage: Legal System has the lowest mean of the five components, which suggests that strong institutional quality is harder to achieve than monetary stability or trade openness. Size of Government has the widest spread, which is a preview of the heterogeneity problem the paper comes back to later.

```
merged_data %>%
  select(log_gdp_pc, efw_total, efw_gov,
         efw_legal, efw_money, efw_trade, efw_reg) %>%
  summary()
```

##	log_gdp_pc	efw_total	efw_gov	efw_legal
##	Min. : 5.493	Min. :2.562	Min. : 2.088	Min. :1.178
##	1st Qu.: 7.313	1st Qu.:5.852	1st Qu.: 5.898	1st Qu.:3.883
##	Median : 8.477	Median :6.609	Median : 6.687	Median :5.037
##	Mean : 8.547	Mean :6.549	Mean : 6.660	Mean :5.230
##	3rd Qu.: 9.769	3rd Qu.:7.401	3rd Qu.: 7.524	3rd Qu.:6.419
##	Max. :11.630	Max. :9.209	Max. :10.000	Max. :9.243
##				NA's :4
##	efw_money	efw_trade	efw_reg	
##	Min. : 0.00807	Min. :1.765	Min. :2.249	
##	1st Qu.: 6.56279	1st Qu.:6.228	1st Qu.:5.446	
##	Median : 7.89642	Median :7.212	Median :6.317	
##	Mean : 7.57312	Mean :7.150	Mean :6.207	
##	3rd Qu.: 9.06185	3rd Qu.:8.269	3rd Qu.:6.982	
##	Max. :10.00000	Max. :9.876	Max. :9.257	
##		NA's :310		

The average overall EFW score sits around 6.5 out of 10, roughly in the middle of the scale. The index is designed to pick up meaningful variation rather than push countries to either extreme, so that's about what you'd expect. The spread across components is more interesting. Size of Government actually posts the highest average of the five, which is a little surprising given how it behaves in the correlation and regression analysis later. Legal System is the lowest, and not by a small margin. Establishing reliable courts, enforceable property rights, and a genuinely independent judiciary turns out to be one of the hardest governance problems there is, and the data reflects that. Most countries manage reasonably well on monetary policy and trade openness long before they get institutions that actually protect the rule of law in a consistent way.

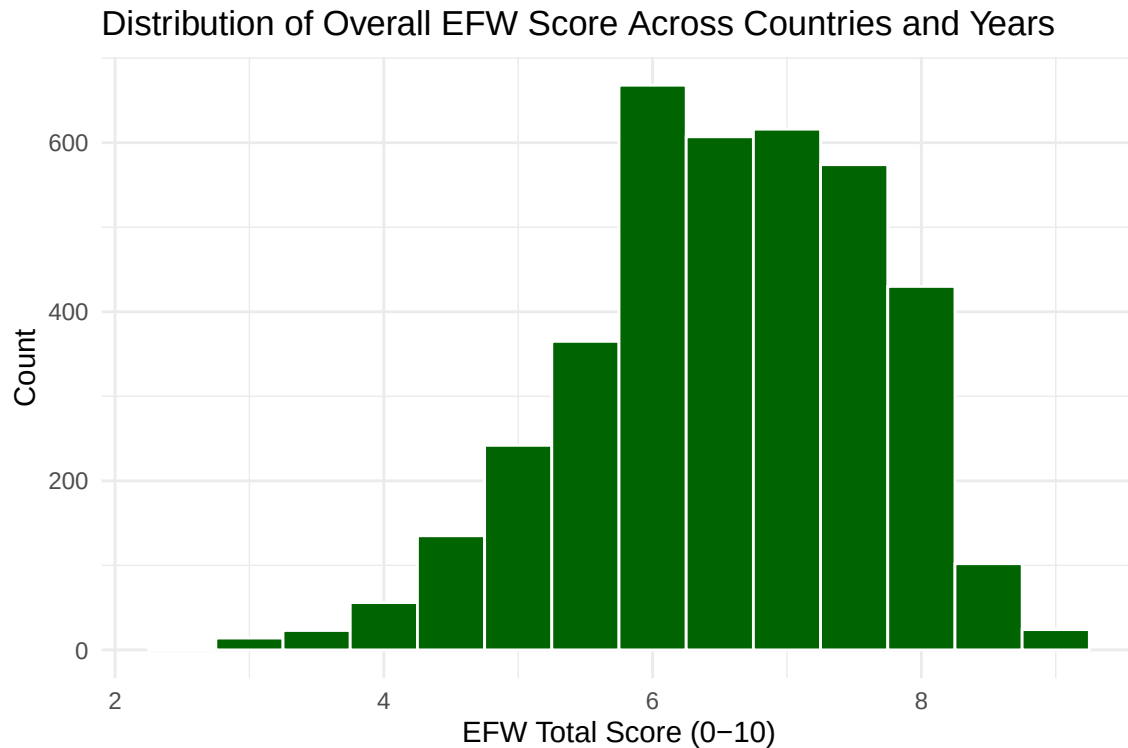
Exploratory Data Analysis

Descriptive Visualizations

Before running any regressions, it's worth just looking at the data. Regression can tell you whether a relationship clears a significance threshold, but it doesn't replace the intuition you get from seeing how the variables are actually distributed and how they move over time. The following plots cover three things: the overall distribution of EFW scores, how scores break down across income groups, and how each component has trended over the full sample period.

```
# Distribution of overall EFW score across all countries and years
ggplot(merged_data, aes(x = efw_total)) +
  geom_histogram(binwidth = 0.5, fill = "darkgreen", color = "white") +
```

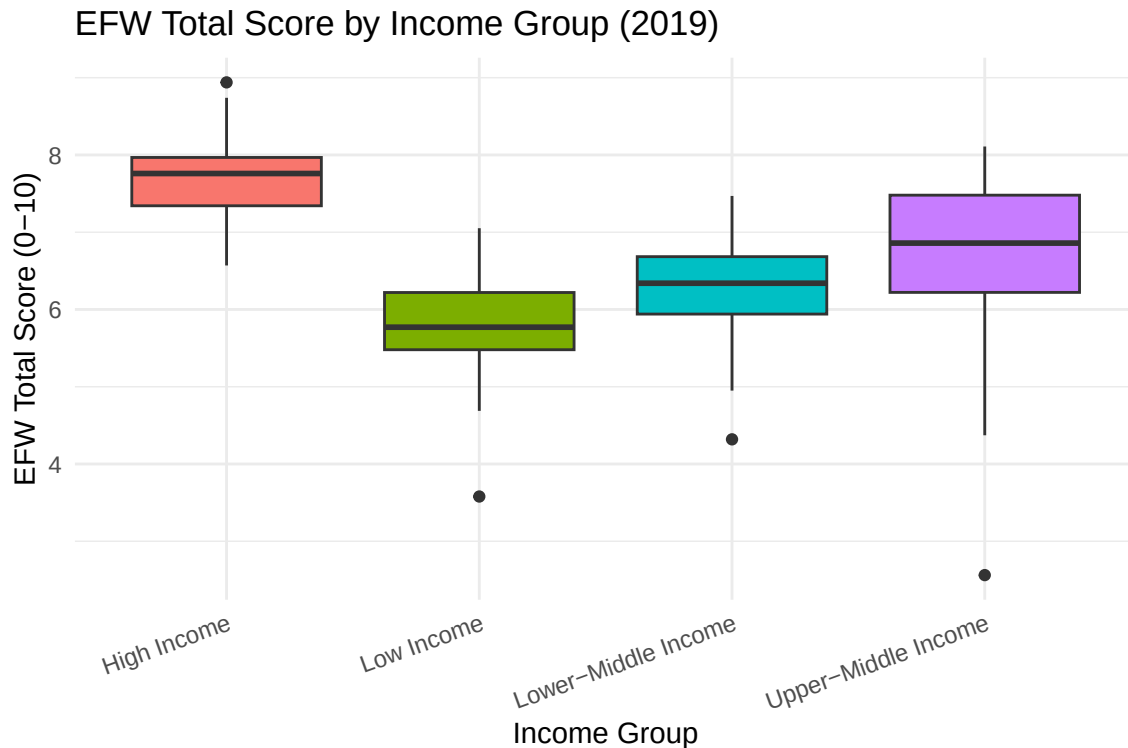
```
labs(
  title = "Distribution of Overall EFW Score Across Countries and Years",
  x = "EFW Total Score (0-10)",
  y = "Count"
) +
theme_minimal()
```



The distribution of overall EFW scores is roughly bell-shaped, centered around 6.5, with most countries landing in the moderate range rather than at either end. Very few score below 3 or above 9, so genuinely unfree economies and near-perfectly free ones are both uncommon. Most of the action is in the middle, which is where the interesting variation actually is.

```
# Create 2019 cross section for plots requiring a single year
cross_section <- merged_data %>%
  filter(year == 2019)

# EFW total score by income group
cross_section %>%
  filter(!is.na(income)) %>%
  ggplot(aes(x = income, y = efw_total, fill = income)) +
  geom_boxplot() +
  labs(
    title = "EFW Total Score by Income Group (2019)",
    x = "Income Group",
    y = "EFW Total Score (0-10)"
  ) +
  theme_minimal() +
  theme(legend.position = "none",
        axis.text.x = element_text(angle = 20, hjust = 1))
```

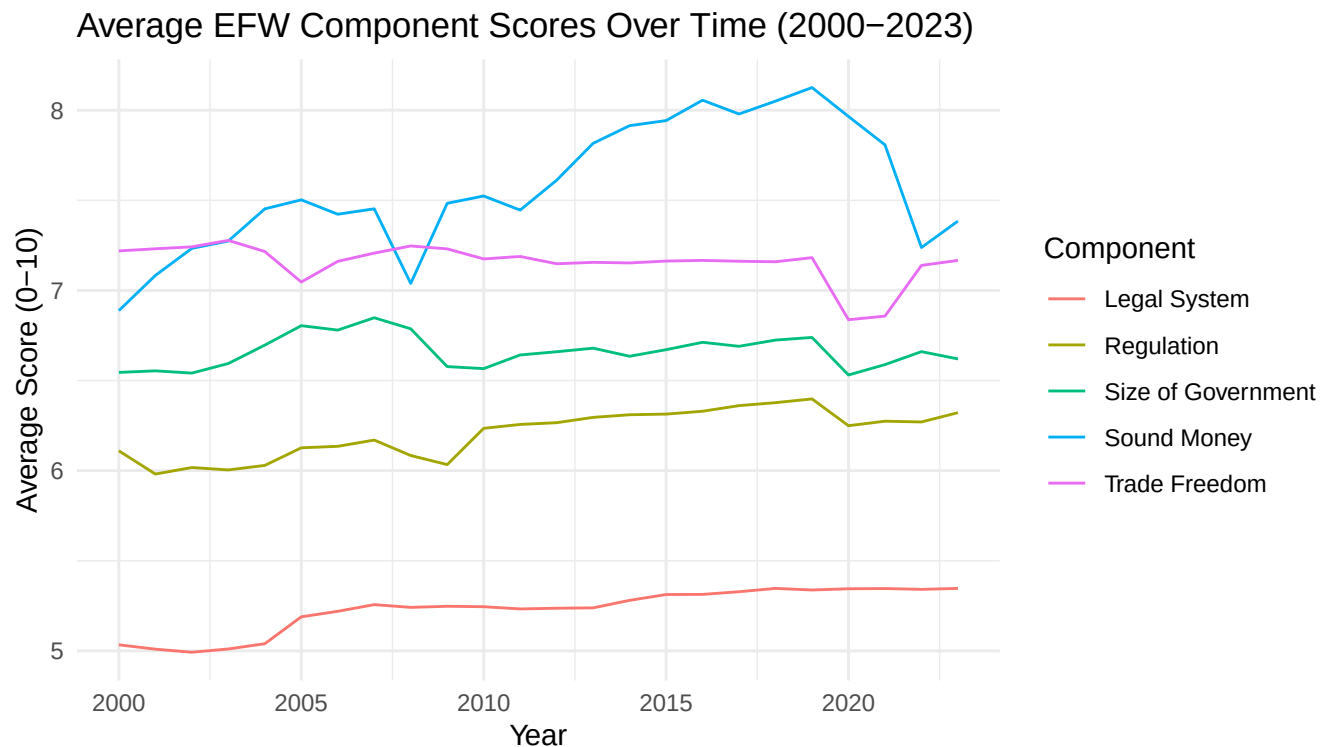


Richer countries score higher on economic freedom. This pattern comes through clearly in the box plot and is consistent with the broader literature. High income countries post a higher median and considerably less spread than low income countries, meaning they not only do better on average but do so more consistently. That said, this is a descriptive relationship. It doesn't tell you whether economic freedom is what made those countries wealthy, or whether being wealthy made it easier to build the institutions that produce economic freedom. That is the question the regression aims to tackle.

```
# Calculate average score for each component by year
avg_by_year <- merged_data %>%
  group_by(year) %>%
  summarize(
    Gov  = mean(efw_gov,   na.rm = TRUE),
    Legal = mean(efw_legal, na.rm = TRUE),
    Money = mean(efw_money, na.rm = TRUE),
    Trade = mean(efw_trade, na.rm = TRUE),
    Reg   = mean(efw_reg,   na.rm = TRUE)
  )

# Plot each component as its own line
ggplot(avg_by_year, aes(x = year)) +
  geom_line(aes(y = Gov,   color = "Size of Government")) +
  geom_line(aes(y = Legal, color = "Legal System")) +
  geom_line(aes(y = Money, color = "Sound Money")) +
  geom_line(aes(y = Trade, color = "Trade Freedom")) +
  geom_line(aes(y = Reg,   color = "Regulation")) +
  labs(
    title = "Average EFW Component Scores Over Time (2000-2023)",
    x      = "Year",
    y      = "Average Score (0-10)",
    color  = "Component"
  )
```

```
) +  
theme_minimal()
```



The time trends tell an interesting story. Trade Freedom and Sound Money show the most visible movement across the 23-year period. Trade Freedom dipped around the 2008 financial crisis and again during COVID, which are both moments when countries pulled inward and trade barriers went up. Sound Money improved steadily through the 2000s as central banks got inflation under control, then reversed in recent years as post-pandemic inflation came back. Size of Government barely moves. It sits roughly flat across the entire period, which turns out to be consistent with what the correlation analysis finds later: it doesn't just correlate weakly with the other components at a point in time, it also fails to respond to the same global forces that visibly shift them over time.

Correlations Between EFW Components and Overall Economic Freedom

The central question here is which components actually drive overall economic freedom. The most direct way to start answering that is to look at how closely each component moves with the total EFW score. Correlation is the right tool since it measures the strength of a linear relationship on a scale from 0 to 1, where values near 1 mean the two variables move tightly together and values near 0 mean they're essentially unrelated. The expectation going in was that all five would score high — the Fraser index weights them equally, and the design assumption built into that is that each one contributes roughly the same amount to the overall score.

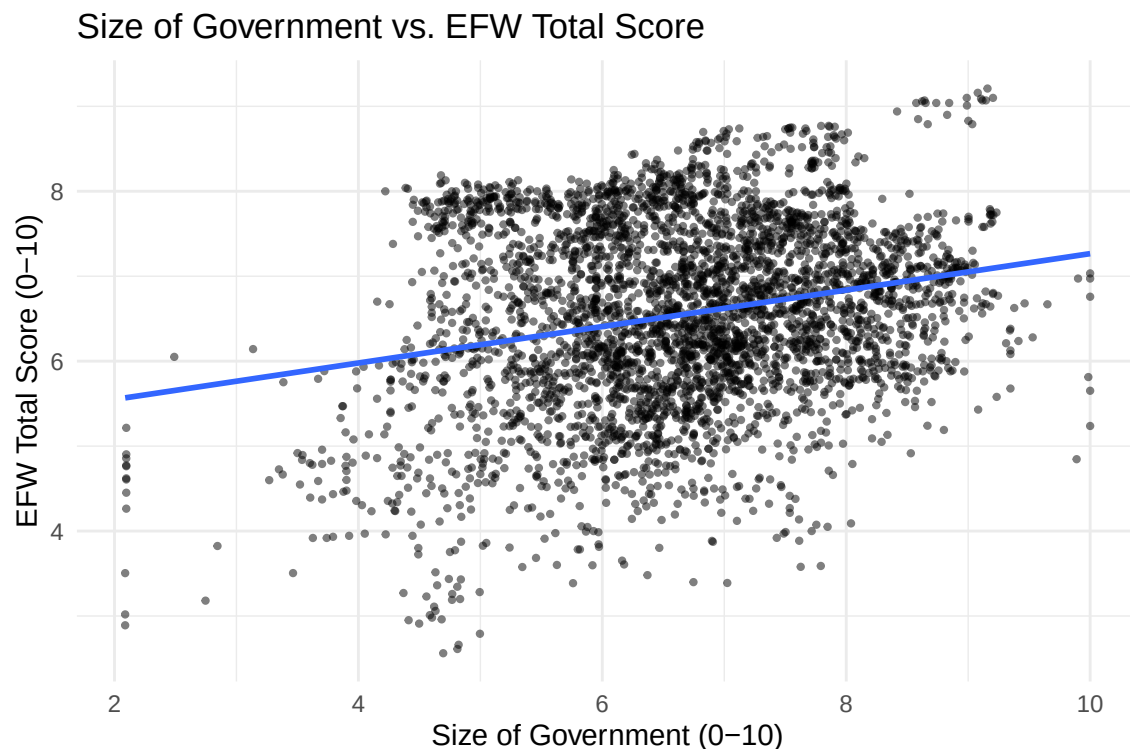
```
merged_data %>%  
  summarize(  
    r_gov = cor(merged_data$efw_gov, merged_data$efw_total, use = "complete.obs"),  
    r_legal = cor(merged_data$efw_legal, merged_data$efw_total, use = "complete.obs"),  
    r_money = cor(merged_data$efw_money, merged_data$efw_total, use = "complete.obs"),  
    r_trade = cor(merged_data$efw_trade, merged_data$efw_total, use = "complete.obs"),  
    r_reg = cor(merged_data$efw_reg, merged_data$efw_total, use = "complete.obs")  
  )
```

```
## # A tibble: 1 x 5
##   r_gov r_legal r_money r_trade r_reg
##   <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 0.237  0.825  0.826  0.886 0.861
```

The results don't support that assumption. Trade Freedom (0.89), Regulation (0.86), Sound Money (0.83), and Legal System (0.82) all correlate strongly with overall economic freedom — tight, consistent relationships that hold up across the full dataset. Then there's Size of Government at 0.24, which shows almost no relationship at all. Four of the five components behave exactly the way the index design assumes they should. The fifth behaves like it's measuring something else entirely. This is the paper's central finding, and it holds up through every subsequent analysis.

The scatter plots below put a face on those numbers. Each point is one country-year observation, and there are nearly 3,900 of them across five charts.

```
ggplot(merged_data, aes(x = efw_gov, y = efw_total)) +
  geom_point(size = 0.8, alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Size of Government vs. EFW Total Score",
       x = "Size of Government (0-10)",
       y = "EFW Total Score (0-10)") +
  theme_minimal()
```

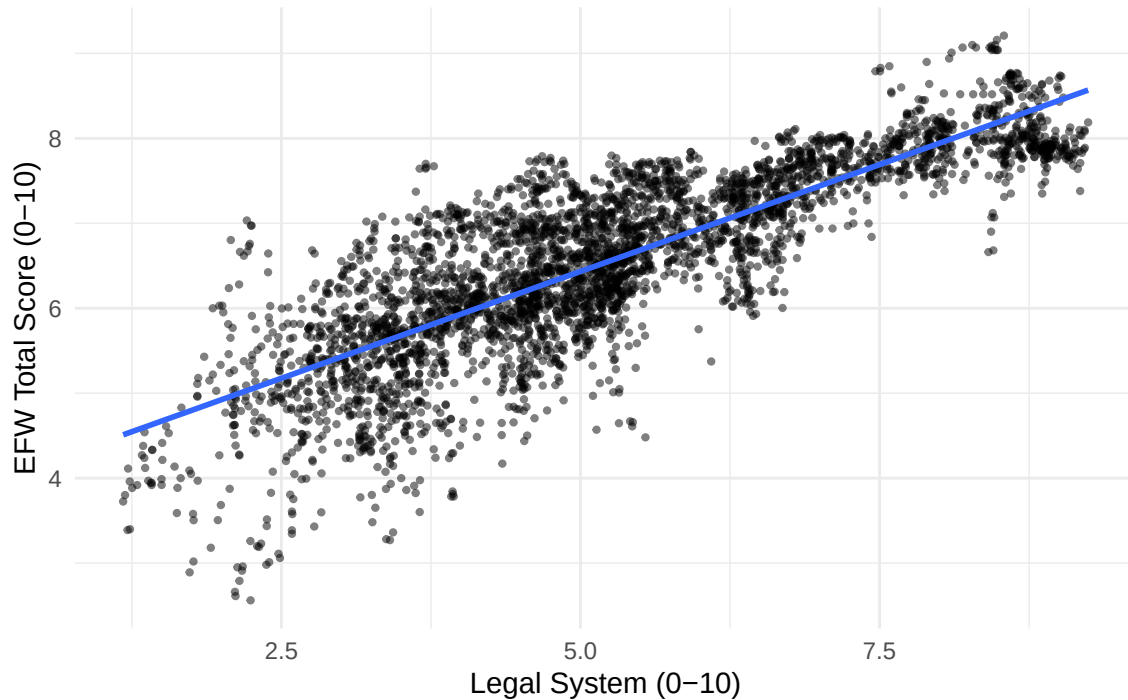


```
ggplot(merged_data, aes(x = efw_legal, y = efw_total)) +
  geom_point(size = 0.8, alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Legal System vs. EFW Total Score",
       x = "Legal System (0-10)",
       y = "EFW Total Score (0-10)") +
  theme_minimal()
```

```
## Warning: Removed 4 rows containing non-finite outside the scale range
## (`stat_smooth()`).

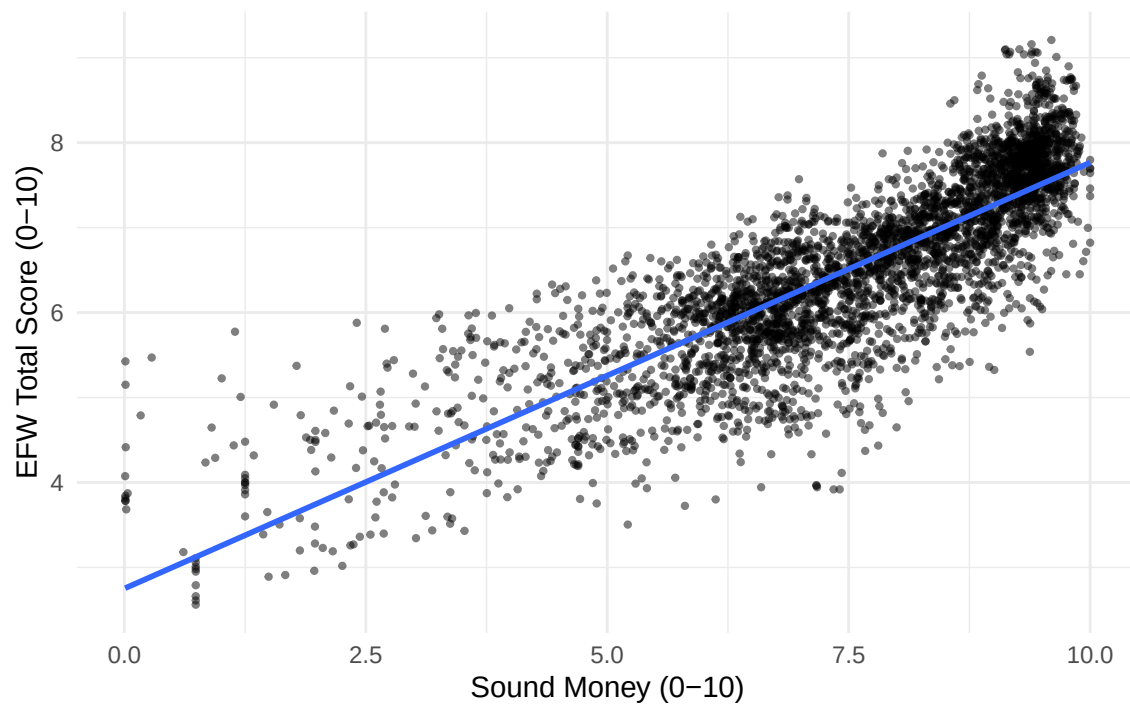
## Warning: Removed 4 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

Legal System vs. EFW Total Score



```
ggplot(merged_data, aes(x = efw_money, y = efw_total)) +
  geom_point(size = 0.8, alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Sound Money vs. EFW Total Score",
       x = "Sound Money (0-10)",
       y = "EFW Total Score (0-10)") +
  theme_minimal()
```

Sound Money vs. EFW Total Score

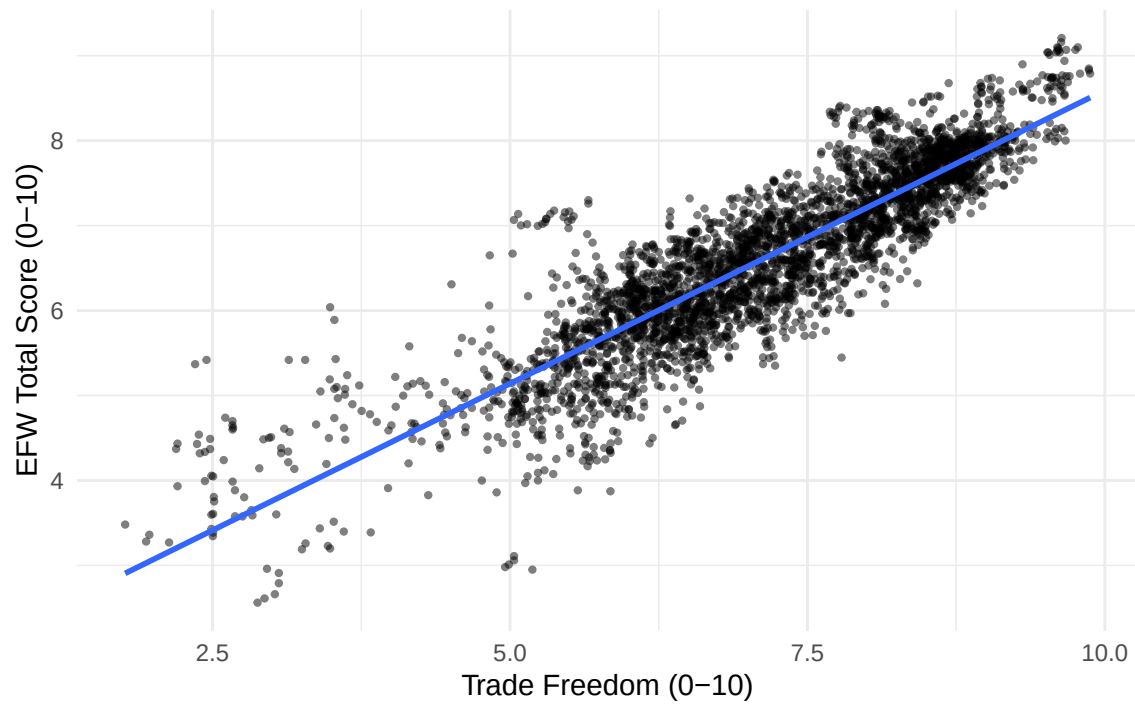


```
ggplot(merged_data, aes(x = efw_trade, y = efw_total)) +  
  geom_point(size = 0.8, alpha = 0.5) +  
  geom_smooth(method = "lm", se = FALSE) +  
  labs(title = "Trade Freedom vs. EFW Total Score",  
        x = "Trade Freedom (0-10)",  
        y = "EFW Total Score (0-10)") +  
  theme_minimal()
```

```
## Warning: Removed 310 rows containing non-finite outside the scale range  
## (`stat_smooth()`).
```

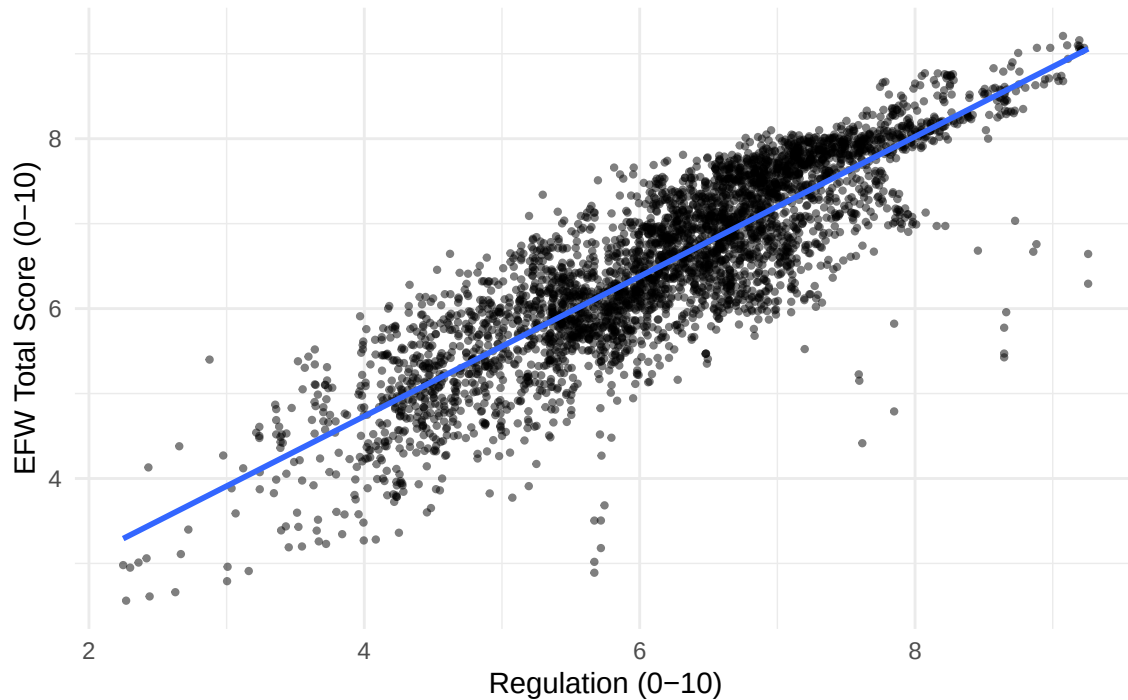
```
## Warning: Removed 310 rows containing missing values or values outside the scale range  
## (`geom_point()`).
```

Trade Freedom vs. EFW Total Score



```
ggplot(merged_data, aes(x = efw_reg, y = efw_total)) +  
  geom_point(size = 0.8, alpha = 0.5) +  
  geom_smooth(method = "lm", se = FALSE) +  
  labs(title = "Regulation vs. EFW Total Score",  
        x = "Regulation (0-10)",  
        y = "EFW Total Score (0-10)") +  
  theme_minimal()
```

Regulation vs. EFW Total Score



The difference between the Size of Government scatter plot and the other four is immediate. Trade Freedom, Regulation, Sound Money, and Legal System all show tight upward slopes, with points that cluster close to the line. This is a clear and consistent signal that higher scores on those components positively correlate with higher overall economic freedom. The Size of Government plot is a cloud. Points scatter in every direction with no discernible pattern, which means a country with a low Size of Government score is about as likely to be economically free overall as one with a high score. The component just doesn't predict where a country falls on the index. The scatter plots are saying the same thing the correlation coefficients say, just more visually.

Regression Methodology

To formally test which EFW components predict prosperity, the analysis uses two regression models. The first is a straightforward OLS regression on the 2019 cross-section, with one observation per country and 163 countries total. This captures the cross-sectional relationship between each policy dimension and GDP per capita at a single point in time. 2019 was chosen specifically because it is the last clean pre-COVID year before the pandemic introduced economic disruptions that had nothing to do with the policy variables being studied. The second model uses the full 2000–2023 panel and adds country and year fixed effects, which is a more demanding setup that controls for a much wider range of confounding factors.

Country fixed effects work by comparing each country to itself across time rather than comparing different countries to one another. Instead of asking whether the United States is richer than Nigeria, a comparison contaminated by centuries of different history, such as geography, and institutional development. The fixed effects model asks whether a given country's GDP tends to be higher in years when, say, its Trade Freedom score is higher. That within-country comparison strips out everything that makes countries permanently different from each other: geography, colonial history, culture, legal traditions. Year fixed effects handle a separate problem, namely global events that hit all countries at once. A worldwide recession that tanks GDP everywhere could otherwise look like a shift in economic freedom policy. The comparison between the two

models works as a robustness check: if the results hold across both specifications, they're harder to dismiss as an artifact of which countries happen to be rich.

Log GDP per capita is the dependent variable throughout. The reason for logging it is practical. Raw GDP per capita ranges from a few hundred dollars in the poorest countries to over seventy thousand in the richest, and that kind of range makes the raw relationship with policy variables highly nonlinear, which is hard to model cleanly with OLS. Taking the natural log compresses that range and brings the relationship much closer to linear, which is what OLS assumes. It also makes the coefficients easier to interpret: a one-unit increase in an EFW component maps to an approximate percentage change in GDP per capita rather than a dollar figure, which makes more sense when you're comparing countries at completely different income levels. Log-transforming income is common practice in growth economics for exactly these reasons.

Results

Regression Models

```
# Model 1: Simple OLS - all 5 EFW components predicting log GDP per capita
# 2019 cross-section, no fixed effects
model_simple <- lm(log_gdp_pc ~ efw_gov + efw_legal + efw_money +
  efw_trade + efw_reg, data = cross_section)

tidy(model_simple, conf.int = TRUE)
```

```
## # A tibble: 6 x 7
##   term          estimate std.error statistic  p.value conf.low conf.high
##   <chr>          <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 (Intercept)    5.61      0.554    10.1    6.95e-19  4.52      6.71
## 2 efw_gov       -0.214     0.0635   -3.36   9.74e- 4 -0.339    -0.0881
## 3 efw_legal      0.497     0.0711    7.00   7.15e-11  0.357     0.638
## 4 efw_money      0.0525    0.0586    0.896  3.71e- 1 -0.0632    0.168
## 5 efw_trade      0.153     0.0811    1.88   6.14e- 2 -0.00742   0.313
## 6 efw_reg        0.0528    0.116     0.456  6.49e- 1 -0.176     0.282
```

```
glance(model_simple)
```

```
## # A tibble: 1 x 12
##   r.squared adj.r.squared sigma statistic  p.value    df logLik   AIC   BIC
##   <dbl>      <dbl> <dbl>    <dbl>    <dbl> <dbl> <dbl> <dbl> <dbl>
## 1    0.677      0.667 0.828     65.9 8.13e-37     5 -197.  409.  431.
## # i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

```
# Model 2: Fixed effects - country and year fixed effects, full panel 2000-2023
```

```
# factor(iso_code) creates a dummy variable for each country.
# This controls for all time-invariant characteristics of each country
# such as geography, culture, and historical institutions,
# by comparing each country to itself over time
# rather than comparing different countries to each other.

# factor(year) creates a dummy variable for each year.
# This controls for global shocks that affect all countries simultaneously
# such as the 2008 financial crisis or commodity price swings.
```

```
model_fe <- lm(log_gdp_pc ~ efw_gov + efw_legal + efw_money +
```

```

efw_trade + efw_reg + factor(iso_code) + factor(year),
data = merged_data)

tidy(model_fe, conf.int = TRUE)

## # A tibble: 192 x 7
##   term                estimate std.error statistic  p.value conf.low conf.high
##   <chr>                <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 (Intercept)          7.07      0.0582    121.    0          6.96      7.18
## 2 efw_gov             -0.0127   0.00487   -2.60 9.28e- 3 -0.0222 -0.00313
## 3 efw_legal            0.0547   0.00944    5.79 7.84e- 9  0.0361  0.0732
## 4 efw_money            0.0253   0.00294    8.59 1.32e- 17 0.0195  0.0310
## 5 efw_trade            0.0370   0.00491    7.53 6.48e- 14 0.0274  0.0466
## 6 efw_reg              0.0902   0.00753   12.0 1.98e- 32 0.0754  0.105
## 7 factor(iso_code)ALB -0.401    0.0486   -8.26 2.13e- 16 -0.496 -0.306
## 8 factor(iso_code)ARE  2.14     0.0484   44.2 0          2.04    2.23
## 9 factor(iso_code)ARG  1.10     0.0447   24.7 4.95e-124  1.02    1.19
## 10 factor(iso_code)ARM -0.442    0.0490   -9.02 3.22e- 19 -0.538 -0.346
## # i 182 more rows

```

```
glance(model_fe)
```

```

## # A tibble: 1 x 12
##   r.squared adj.r.squared sigma statistic p.value    df logLik    AIC    BIC
##   <dbl>      <dbl> <dbl>    <dbl>    <dbl> <dbl> <dbl>    <dbl>
## 1    0.992    0.992 0.132    2271.    0    191  2255. -4125. -2933.
## # i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>

```

```

# coef_omit = "factor" hides the country and year dummy coefficients
# from the output table to keep it readable.
# There are approximately 160 country dummies and 23 year dummies.
# We only care about the five EFW component coefficients.

```

```

modelsummary(
  list(
    "Simple OLS (2019 Cross-Section)" = model_simple,
    "Fixed Effects (Full Panel)"      = model_fe
  ),
  stars      = TRUE,
  statistic = "std.error",
  coef_omit = "factor",
  notes     = "Standard errors in parentheses. Model 1: 2019 cross-section. Model 2: full 2000-2023 panel"
)

```

Interpretation of Regression Results

Simple OLS (2019 Cross-Section) In the 2019 cross-section, the clearest story comes from two variables. On average, a one unit increase in Legal System score is associated with approximately 49.7% higher GDP per capita, holding the other four components constant. This is the largest positive effect in the model and significant at $p < 0.001$. Size of Government moves in the opposite direction: a one unit increase is associated with approximately 21.4% lower GDP per capita on average, also holding all other components constant, also significant at $p < 0.001$. Trade Freedom shows a positive association of about 15.3% but only clears the 10% significance threshold, so it should be interpreted cautiously. Sound Money and Regulation both point positive at around 5.3% each, but neither reaches statistical significance in this single-year snapshot. This is likely because the cross-section is too noisy to isolate their effects from everything else that differs between

	Simple OLS (2019 Cross-Section)	Fixed Effects (Full Panel)
(Intercept)	5.614*** (0.554)	7.070*** (0.058)
efw_gov	-0.214*** (0.064)	-0.013** (0.005)
efw_legal	0.497*** (0.071)	0.055*** (0.009)
efw_money	0.053 (0.059)	0.025*** (0.003)
efw_trade	0.153+ (0.081)	0.037*** (0.005)
efw_reg	0.053 (0.116)	0.090*** (0.008)
Num.Obs.	163	3545
R2	0.677	0.992
R2 Adj.	0.667	0.992
AIC	408.9	-4124.9
BIC	430.5	-2933.5
Log.Lik.	-197.429	2255.462
RMSE	0.81	0.13

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Standard errors in parentheses. Model 1: 2019 cross-section. Model 2: full 2000-2023 panel with country and year fixed effects.

countries in 2019.

Fixed Effects (Full Panel) The fixed effects panel tells a more complete story. Once permanent country differences and global time trends are controlled for, all five components become statistically significant. This includes Sound Money, Trade Freedom, and Regulation, which were indistinguishable from noise in the cross-section. Regulation shows the largest within-country effect at approximately 9.0% per unit increase on average, holding all other components constant. Legal System comes in at 5.5%, Trade Freedom at 3.7%, and Sound Money at 2.5%, all significant at $p < 0.001$. Size of Government remains negative at -1.3% and significant at $p < 0.01$. The fact that the direction of every coefficient is consistent across both models, even as the magnitudes shift, is reassuring, since the story the cross-section tells is not contradicted by the panel, rather it is refined by it.

Discussion

The regression results do two things. They confirm what the correlation analysis suggested: that four of the five components are meaningful predictors of economic outcomes and they extend that finding by connecting it directly to GDP per capita rather than just the index score itself. If Size of Government had been weak in the correlations but strong in the GDP regression, that would complicate the interpretation. The fact that it underperforms in both analyses points to something more substantive.

The correlation analysis showed that Trade Freedom, Regulation, Sound Money, and Legal System all move closely with overall economic freedom, while Size of Government barely moves with it at all. The regression confirms that the same pattern shows up when predicting actual GDP per capita. That consistency across two very different analytical approaches makes the finding harder to dismiss as a quirk of how the Fraser index is put together.

The explanation for why Size of Government behaves so differently comes down to what it actually measures. The Fraser index captures government size as an aggregate, meaning spending as a share of GDP, tax rates, state-owned enterprises. It does so without distinguishing between what that spending is actually doing. Two countries can post identical Size of Government scores while operating completely different economic environments. One might be directing resources toward infrastructure, courts, and public goods that enable market activity. Another might be funding transfer payments, subsidies, and state enterprises that crowd out private investment. The Fraser Institute's aggregate score treats both of these things identically.

A useful analogy would be to imagine a health study that uses total food spending as a proxy for dietary quality, on the theory that healthier food costs more. The hypothesis seems reasonable going in, but the correlation ends up weak. This is because the dollar figure doesn't tell you whether someone is buying organic produce or fast food. The same logic applies here. Aggregate government spending doesn't tell you whether the money is going toward things that build an economy or things that distort it.

Singapore makes this concrete. It consistently ranks in the top three countries globally on overall economic freedom, yet carries substantial government debt and meaningful public sector involvement in the economy. The reason is that Singapore's spending is overwhelmingly directed toward productive ends, such as infrastructure, education, sovereign wealth, all of which generate real returns. The Netherlands offers a similar example historically: a significant share of government expenditure went toward water management infrastructure that has kept the country habitable and productive for centuries. An aggregate spending measure has no way of distinguishing either of these from wasteful expenditure elsewhere.

What this points toward is a research agenda focused on composition rather than size. The meaningful question is not how much governments spend but what they spend it on, and the data needed to answer that question at a cross-country scale does not yet exist in a form suitable for this kind of analysis.

Limitations

The most important limitation is that nothing in this analysis establishes causation. The correlations and regressions identify associations but they cannot rule out the possibility that wealthier countries simply develop better institutions over time rather than better institutions making countries wealthier. This reverse causality problem is endemic to cross-country observational work, and this paper is not immune to it.

Second, as discussed at length above, the Fraser measure of government size is an aggregate that cannot distinguish productive from unproductive spending. This is almost certainly the main driver of its anomalous behavior in both analyses, but the data available here do not allow a direct test of that hypothesis. It remains an informed inference rather than a demonstrated fact.

Third, fixed effects are a powerful tool but not a complete solution to omitted variable bias. They handle everything that doesn't change within a country over time, such as geography, colonial history, and deep cultural factors. But time-varying omitted variables still remain a concern. Education levels, political stability, and natural resource discoveries mid-sample could all be influencing both the EFW scores and GDP in ways this model doesn't fully account for.

Finally, GDP per capita is measured here in constant 2015 US dollars rather than purchasing power parity adjusted figures. This affects level comparisons across countries (a dollar goes further in some places than others), though it is less likely to change the direction of the relationships estimated in the regression.

Making this analysis causal rather than correlational is the most important direction for future work. The gold standard in this literature would be a natural experiment: a setting where countries were effectively assigned to different economic policy environments by something outside their control rather than by their own choices. The partition of Germany after World War II and the division of Korea are the canonical examples: identical populations, shared histories, suddenly subjected to completely different economic systems. Comparing outcomes across those divides gets much closer to a genuine causal estimate because the assignment was not driven by the countries choosing their own policies. Short of a natural experiment, two econometric approaches could strengthen the causal interpretation of these findings. A differences-in-differences design would identify countries that underwent major policy reforms, such as a significant liberalization of trade rules, an independence of the central bank, or a judicial overhaul. It would then compare their trajectories before and after the reform against countries that did not experience similar changes. Moreover, an instrumental variables approach would require finding something that predicts a country's EFW component scores but affects GDP only through that channel and not through any other route. This is a difficult condition to satisfy in cross-country work, but it is not impossible. Until one of those designs can be implemented with comparable data coverage, the relationships documented here should be interpreted as robust associations rather than causal effects.

Conclusion

Four of the five Fraser EFW components (Trade Freedom ($r = 0.89$), Regulation ($r = 0.86$), Sound Money ($r = 0.83$), and Legal System ($r = 0.82$)) are strong predictors of overall economic freedom and consistently associated with higher GDP per capita. Size of Government is the clear outlier, with a correlation of just 0.24 with the overall index and a negative coefficient in both regression models.

The regression analysis bears this out since Legal System, Sound Money, Trade Freedom, and Regulation are statistically significant in both specifications. Size of Government comes out negative in both, a finding that survives controlling for permanent country differences and global time trends. Its poor performance isn't surprising given what the measure actually captures, but the consistency of the result across both models makes it hard to set aside.

Countries that protect the rules of the game, through honest courts, stable money, open trade, and sensible regulation, tend to score high on economic freedom and tend to have higher per capita incomes. How much a government spends, taken by itself, tells you very little about either. The more interesting question for future

research is what governments actually spend their money on, and whether a measure that captures spending composition rather than spending volume would prove a stronger predictor of both economic freedom and prosperity than anything currently available in cross-country indices.

References

- Gwartney, J., Lawson, R., and Block, W. (1996). *Economic Freedom of the World: 1975–1995*. Fraser Institute.
- Gwartney, J., Lawson, R., and Murphy, R. (2024). *Economic Freedom of the World: 2024 Annual Report*. Fraser Institute.
- Heckelman, J.C. and Stroup, M.D. (2000). Which economic freedoms contribute to growth? *Kyklos*, 53(4), 527–544.
- Heckelman, J.C. and Stroup, M.D. (2005). A comparison of aggregation methods for measures of economic freedom. *European Journal of Political Economy*, 21(4), 953–966.